

SA + PBIL: Coupling Simulated Annealing with Population-Based Incremental Learning for Discrete Choice Model Specification Search

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Motivation

The search for the best specification of a discrete choice model involves navigating a high-dimensional combinatorial space where each candidate specification is defined by a large number of simultaneous modelling decisions. The space grows exponentially with the number of candidate variables, and evaluating each specification requires estimating a full model via maximum simulated likelihood, which is computationally expensive. A purely exhaustive search is therefore infeasible, and a metaheuristic algorithm is needed. The framework described here is general and applies to any discrete choice model where the specification decisions take the form described below. It has currently been implemented and tested for Mixed Logit (MXL) and Mixed Random Regret Minimisation (MXRRM) models, with extensions to other model families planned for future work.

Simulated Annealing (SA) is a natural candidate for this problem given its well-established ability to escape local optima through a probabilistic acceptance criterion controlled by a temperature parameter. However, SA alone carries no memory between iterations: each candidate specification is generated by perturbing the current accepted solution, but the perturbation process is entirely uninformed by the history of the search. During early search runs with SA alone, an empirical pattern was observed: certain variables appeared consistently significant across many accepted specifications, while others were rarely significant regardless of the combination in which they appeared. This does not mean the search space has a fixed structure of this kind, significance is context-dependent and a variable that is insignificant in one combination may be significant in another, but it suggests that recurring patterns may emerge during the search, and that an algorithm capable of recognising and exploiting those patterns would be more efficient than one that ignores them. This observation motivated the integration of Population-Based Incremental Learning (PBIL) into the search, giving rise to the SA+PBIL algorithm described here.

The Probability Matrix

PBIL maintains a probability matrix that encodes what the algorithm has learned so far about the structure of good specifications. Before describing how the matrix is updated and used, it is important to note that the analyst may pre-specify certain modelling decisions on substantive grounds, for example mandating that a particular variable must always be included, or that a given coefficient must be treated as random. These pre-specifications are hard constraints that override the probability matrix entirely and are always respected regardless of what the statistical evidence accumulated during the search would suggest.

For all remaining decisions not pre-specified by the analyst, the probability matrix maintains one entry per candidate variable, and for each variable it tracks five decisions in a hierarchical and conditional manner. The first decision is inclusion: whether the variable enters the model at all. If included, the second decision is whether its coefficient should be treated as random across individuals or fixed. If random, the third decision is which mixing distribution to assign, chosen from a candidate set comprising normal (n), lognormal (ln), truncated normal (tn), uniform (u), and triangular (t)

distributions. The fourth decision is whether the random coefficient should be correlated with at least one other random coefficient in the model. The fifth decision is whether a Box-Cox transformation should be applied to the variable. These decisions are not independent and are subject to the following constraints: correlation requires the coefficient to be random, and correlation and Box-Cox transformation are mutually exclusive for the same variable. These constraints are enforced during sampling so that every candidate specification generated by the algorithm is always structurally valid.

In the current implementation, all probabilities are initialised at 0.5 for scalar decisions, representing maximum prior uncertainty. For the distribution decision, probabilities are initialised uniformly across all candidate distributions, so each distribution starts with equal probability $1/|F|$ where $|F|$ is the number of candidate distributions. All probabilities are clamped to the interval $[0.05, 0.95]$ at all times, which prevents any modelling decision from being permanently excluded from or locked into the search regardless of accumulated evidence.

This initialisation makes the difference between SA alone and SA+PBIL clear. At the very start of the search the probability matrix is uniform and the behaviour of SA+PBIL is nearly indistinguishable from SA alone, since no evidence has yet been accumulated and every modelling decision is equally likely. As the search progresses and more solutions are accepted, the matrix gradually departs from uniformity and begins to reflect the patterns observed in good specifications. The perturbation and sampling process therefore becomes increasingly guided over time, concentrating effort on promising regions of the specification space. This progressive guidance is the fundamental difference between the two approaches.

Learning from Accepted Solutions

The probability matrix is updated every time SA accepts a new solution. The choice to update on accepted solutions rather than on all evaluated solutions is deliberate: rejected solutions come from regions of the search space that the acceptance criterion has identified as insufficiently promising, and using them to update the probability model would introduce noise from bad specifications rather than signal from good ones. The choice to update on all accepted solutions rather than only on improvements to the best solution found so far is equally deliberate: updating only on improvements would make learning too infrequent, particularly at high temperatures when SA accepts many non-improving moves, and would reduce the diversity of modelling decisions that the probability model is exposed to.

When a solution is accepted, the update rule for any probability p in the matrix follows a weighted average:

$$p \leftarrow (1 - l) \cdot p + l \cdot \mathbf{1}[\text{significant}] \quad (1)$$

where $\mathbf{1}[\text{significant}]$ is an indicator equal to 1 if the corresponding coefficient was statistically significant in the accepted solution, and 0 otherwise, and l is the learning rate. This rule gradually shifts each probability toward 1 for decisions associated with statistically significant coefficients and toward 0 for decisions associated with insignificant ones, without discarding past evidence abruptly. The speed at which this shift occurs is controlled by l .

The update rule is not applied uniformly across all decision levels. Each level has its own conditional logic that determines when and how learning occurs, reflecting the hierarchical and gated structure of the probability matrix.

For **inclusion**, the probability increases toward 1 if the mean coefficient of the variable is significant, or if the variable has a random coefficient and its standard deviation is significant. It decreases toward 0 only if the mean is insignificant and either the variable is not random, or both the mean and standard deviation are insignificant simultaneously. This dual condition for the downward update avoids penalising inclusion prematurely when a variable is random and its mean alone happens to be insignificant.

For **randomness**, the update is only triggered when the variable appears as a random coefficient in the accepted solution. The probability increases toward 1 if the standard deviation is significant,

indicating genuine heterogeneity in the population, and decreases toward 0 if the standard deviation is insignificant.

For **distribution type**, the update is also gated on the variable being random in the accepted solution. Learning toward the current distribution occurs only when both the mean and the standard deviation are significant, since both are needed to conclude that the distributional form is well-identified. If only one is significant, the probability of the current distribution is shifted toward 0, reflecting that the distributional assignment is not well-supported in that solution.

For **correlation**, the update is triggered only when the variable is random and does not have a Box-Cox transformation in the accepted solution, consistently with the mutual exclusion constraint. Significance is assessed at the level of the full Cholesky row corresponding to that variable: the variable is considered significantly correlated if at least one of its off-diagonal Cholesky entries with any other correlated variable in the model is statistically significant. The probability increases toward 1 if this row-level criterion is met and decreases toward 0 otherwise. This follows the block-based approach of Ahern et al. (2024), where correlation is treated as a property of the variable’s relationship with the entire correlated block rather than a single pairwise coefficient.

For **Box-Cox transformation**, the update is triggered only when the variable is eligible for Box-Cox and is not correlated in the accepted solution. The probability increases toward 1 if the Box-Cox coefficient is significant and decreases toward 0 otherwise.

Temperature-Coupled Learning Rates

The learning rate l is not fixed. This is the central design decision of the algorithm. A fixed learning rate creates a conflict with the SA temperature schedule: if l is large, PBIL updates the probability matrix aggressively from the beginning of the search, which collapses the diversity of candidate specifications at exactly the stage when SA needs to explore broadly. If l is small, PBIL updates too slowly throughout the entire search and its accumulated knowledge has little influence on the exploitation phase. The solution is to couple the learning rate to the SA temperature so that the rate of learning adapts automatically to the current phase of the search.

The learning rate varies as a function of the current temperature according to:

$$l(T) = l_{\min} + (l_{\max} - l_{\min}) \cdot \left(1 - \frac{T}{T_I}\right) \quad (2)$$

where T is the current temperature, T_I is the initial temperature, and $l_{\min}$ and $l_{\max}$ are level-specific bounds. At the beginning of the search, when $T \approx T_I$, the ratio T/T_I is close to 1 and $l \approx l_{\min}$, so the probability matrix is updated very slowly and SA retains full exploratory freedom. As the temperature decreases and SA transitions from exploration to exploitation, T/T_I approaches 0 and l approaches $l_{\max}$, so updates become faster and the algorithm directs its search more strongly toward the structural patterns it has accumulated. The result is that exploitation in the late stages of the search is not uninformed but guided by a probability model that has been built up gradually over many accepted solutions.

It is worth noting that T_I is calibrated automatically from the data using the procedure of Kirkpatrick et al. (1983), while T_F is defined by the analyst as a stopping criterion. Under the current formulation, the learning rate reaches $l_{\max}$ only asymptotically as $T \rightarrow 0$, not exactly at $T = T_F$. An alternative normalisation using $(T_I - T)/(T_I - T_F)$ would ensure that the learning rate traverses its full range from exactly $l_{\min}$ at the start to exactly $l_{\max}$ at termination, which is arguably more interpretable. This remains an open design choice.

Differentiated Learning Rate Bounds by Decision Level

The learning rate bounds $l_{\min}$ and $l_{\max}$ are differentiated across the five decision levels, because the statistical signal available to guide learning is not equally strong for all decisions. Inclusion carries the strongest signal: the presence or absence of a variable produces large differences in the objective function and the significance of the mean coefficient provides a clear indication of whether the variable belongs in the model. The randomness decision also carries a relatively strong signal, because a

significant standard deviation provides direct evidence of genuine heterogeneity in the population. The distribution type carries the weakest signal, because the difference in model fit between, for example, a normal and a triangular distribution assigned to the same variable is typically small, and convergence to a single distribution type too early in the search would unnecessarily restrict exploration. The specific bounds used at each decision level are as follows:

Decision level	$l_{\min}$	$l_{\max}$
Inclusion	0.02	0.25
Random	0.02	0.15
Distribution	0.01	0.05
Correlation	0.02	0.15
Box-Cox	0.02	0.15

Table 1: Learning rate bounds by decision level.

Generating Candidate Specifications

At each iteration, SA+PBIL generates a candidate specification through a three-step process that combines random selection of what to perturb with PBIL-guided decisions about how to perturb it.

First, SA randomly decides how many perturbations to apply and which dimension to perturb in each one, choosing from inclusion, randomness, distribution, correlation, and Box-Cox. This selection is random and uniform across available dimensions, preserving the exploratory character of SA.

Second, once a dimension is selected, the direction of the perturbation, whether to add or remove a modelling decision at that level, is not random but guided by the current state of the probability matrix. For each dimension, the probability of moving in the addition direction is computed as:

$$p_{\text{add}} = \frac{\bar{p}^{\text{out}}}{\bar{p}^{\text{out}} + (1 - \bar{p}^{\text{in}})} \quad (3)$$

where $\bar{p}^{\text{out}}$ is the mean probability across variables currently outside the solution at that decision level, and $\bar{p}^{\text{in}}$ is the mean probability across variables currently inside the solution at that decision level. When the matrix has accumulated evidence that certain variables belong in the model, $\bar{p}^{\text{out}}$ is high and the algorithm tends to add them. When it has accumulated evidence that variables already in the model do not belong, $1 - \bar{p}^{\text{in}}$ is high and the algorithm tends to remove them. This equation applies analogously for each decision level using its corresponding column in the probability matrix.

Third, the specific variable selected for addition or removal within the chosen dimension is sampled proportionally to its accumulated probability in the matrix, so variables with stronger statistical support are more likely to be selected.

Any dimension pre-specified by the analyst overrides this process entirely. The structural constraints between decisions are enforced after sampling to ensure every candidate specification is always valid.

The Probability Matrix in Practice

To make the above concrete, consider a simplified example with five candidate variables. Var4 has been pre-specified by the analyst as a fixed coefficient, so its Random, distribution, and Correlated entries are N/A from the start. Var1, Var2, and Var3 are eligible for correlation, so their Box-Cox entries are N/A because correlation and Box-Cox are mutually exclusive. Var5 has been pre-specified as non-correlated by the analyst, so its Correlated entry is N/A but Box-Cox remains available. Distribution columns use the abbreviations: n = normal, ln = lognormal, tn = truncated normal, u = uniform, t = triangular.

At initialisation, all free probabilities start at 0.5 for scalar decisions and at $1/5 = 0.20$ for each distribution, representing maximum prior uncertainty:

After many iterations, the matrix has departed substantially from uniformity and reflects the accumulated statistical evidence from accepted solutions:

Variable	Incl.	Rand.	n	ln	tn	u	t	Corr.	BC
Var1	0.50	0.50	0.20	0.20	0.20	0.20	0.20	0.50	N/A
Var2	0.50	0.50	0.20	0.20	0.20	0.20	0.20	0.50	N/A
Var3	0.50	0.50	0.20	0.20	0.20	0.20	0.20	0.50	N/A
Var4	0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.50
Var5	0.50	0.50	0.20	0.20	0.20	0.20	0.20	N/A	0.50

Table 2: Probability matrix at initialisation.

Variable	Incl.	Rand.	n	ln	tn	u	t	Corr.	BC
Var1	0.89	0.82	0.61	0.12	0.15	0.07	0.05	0.78	N/A
Var2	0.12	0.71	0.08	0.55	0.19	0.11	0.07	0.76	N/A
Var3	0.71	0.08	0.20	0.20	0.20	0.20	0.20	0.05	N/A
Var4	0.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.81
Var5	0.83	0.77	0.14	0.09	0.57	0.12	0.08	N/A	0.72

Table 3: Probability matrix after many iterations.

Reading the final state of the matrix: Var1 has appeared consistently in good specifications with a random coefficient, predominantly normal distribution, and strongly correlated with other random coefficients. Var2 rarely enters the model but when it does it tends to be random with a lognormal distribution and correlated, while its low inclusion probability reflects that it adds little in most specifications. Var3 enters frequently but almost always as a fixed coefficient, which is why its distribution probabilities have not moved from uniform since the randomness gate was almost never activated; its low correlation probability reflects the same gate logic. Var4 is pre-specified as fixed and has learned that a Box-Cox transformation is frequently warranted. Var5 enters frequently with a random coefficient following predominantly a truncated normal distribution, and its Box-Cox probability is moderately high since correlation is disabled for this variable.

Empirical Result

The combination of SA with temperature-coupled PBIL consistently finds better model specifications than SA alone, measured by the BIC of the best solution found within the same computational budget. The improvement reflects the fact that in the exploitation phase, the algorithm is not searching blindly but is concentrating probability mass on modelling decisions that have accumulated statistical support across many previous accepted solutions.